PMD Second-Order Effects on Pulse Propagation in Single-Mode Optical Fibers

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***Abstract*—The time-space varying birefringence in single-mode optical fibers causes the polarization mode dispersion (PMD) to be a serious problem in high bit-rate transmissions. The PMD first- and second-order statistics are well known in the literature, but second-order PMD-induced pulse distortions have still to be clarified for sequences of pulses of arbitrary shape. We give, for the first time, the exact PMD time impulse response, up to second order. We show, both numerically and experimentally, that the effect of the rotation of the principal states of polarization (PSP) is to generate power overshoots on the "1" and "0" bit sequences.**

***Index Terms*—Birefringence, polarization, transient response.**

I. Introduction

SINGLE-MODE fibers are actually bimodal, since they can support two polarization modes traveling at different speeds along the fiber. The travel time difference $\Delta\tau$ between the two modes is called differential group delay (DGD) and the induced broadening of the output signal, referred to as polarization mode dispersion (PMD), can degrade the performance of high bit-rate transmission systems [1]. In a first-order PMD approximation, two replicae of the same input pulse are generated at the fiber output, differentially delayed by $\Delta\tau$, which is a Maxwell distributed random variable [2].

Second-order PMD effects have been studied in the literature [1]–[3], but their impact on output pulse distortion has still to be clarified for sequences of pulses of arbitrary shape. In this letter, we give, for the first time, the exact PMD time impulse response up to second order, and describe the different impact on pulse deformation of the two second-order PMD terms, which are the frequency dependence of the DGD and the frequency dependence of the principal states of polarization (PSP's). Knowledge of the PMD channel transfer function is fundamental for evaluating higher order PMD-induced pulse distortions and to determine the most appropriate PMD equalization techniques.

The letter is organized as follows. Section II reviews the PMD theory, stressing the usefulness of considering the two second-order terms as separate. In Section III, we show how to obtain the exact time impulse response of a PMD fiber, up to the second order. Section IV shows, both numerically and experimentally, that the effect of the rotation of the PSP's is to generate power overshoots, well localized on the output sequence, both on the "1" and "0" bit streams.

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II. PMD Theory

Assuming negligible polarization dependent loss (PDL), a linear dispersive fiber is represented by a 2×2 complex transfer matrix of the form[1] $\boldsymbol{T}(\omega) = e^{-\alpha z}e^{-j\overline{\beta}(\omega)z}\boldsymbol{M}(\omega)$, where α is the fiber attenuation, z is the propagation distance, $\overline{\beta}(\omega)$ is the mean propagation constant and $\boldsymbol{M}(\omega)$ is a unitary matrix [4, eq. (2)]. In the three-dimensional (3-D) Poincaré representation, PMD is described by the dispersion vector $\vec{\Omega} = \Delta\tau\hat{s}$, where $\Delta\tau = |\vec{\Omega}|$ and $\hat{s}$ is the direction of one of the two orthogonal eigenvectors of $\boldsymbol{M}(\omega)$, which represent the PSP's of the fiber. In a first-order approximation both $\Delta\tau$ and $\hat{s}$ are frequency independent, i.e., $\vec{\Omega} = \Delta\tau_0\hat{s}_0$, where $\Delta\tau_0$ is the differential time delay between two input signals polarized along the two PSP's of the fiber. In a second-order approximation $\vec{\Omega}(\omega) = \Delta\tau_0\hat{s}_0 + (\Delta\tau'\hat{s}_0 + 2\vec{k}\Delta\tau_0)\omega$, where $\Delta\tau' = \partial\Delta\tau/\partial\omega$ and $2\vec{k} = \partial\hat{s}/\partial\omega$, where the derivatives are evaluated at the central frequency. Second-order effects are represented by the linear DGD frequency dependence $\Delta\tau'$ and by a linear PSP rotation with constant angular rate $|2\vec{k}|$ [1], [5]. In the following we represent $|2\vec{k}|$ simply with $2k$, as usually done in the literature. Note that the two vectors $\Delta\tau'\hat{s}_0$ and $2\vec{k}\Delta\tau_0$ are actually orthogonal vectors in the Poincaré space.

Carrying out a separate analysis for each of the PMD second-order effects is useful for a better understanding of their different impact on pulse transmission. For example, a separate statistic analysis of the two coefficients $2k$ and $\Delta\tau'$ reveals that the second one can be generally neglected relative to the first [6]. Besides, simulations and experiments prove that $2k$ and $\Delta\tau_0$ are correlated [6], [7], i.e., when $\Delta\tau_0$ is small the corresponding $2k$ is large. Due to this correlation, first-order PMD turns out to be the most important contribution in a statistical analysis of PMD-induced system penalties. Nevertheless, it is important to correctly evaluate second-order effects, especially when first-order PMD compensation techniques are utilized. In the following, we will show evidence of PMD-induced pulse envelope deformations when second-order coefficients take deterministic values.

III. Output Pulse Envelope in Presence of PMD

Consider a generic input field $\boldsymbol{x}_{\text{in}}(t) = x_{\text{in}}(t)\boldsymbol{e}_{\text{in}}$, where $x_{\text{in}}(t)$ represents the field complex amplitude, and $\boldsymbol{e}_{\text{in}} =$

[1] In the following, we utilize the low-pass equivalent representation with respect to the central frequency ω_0 of the optical field. Upper case boldface indicates matrices. Lower case boldface indicates vectors.

$[e_{\text{in}_x}, e_{\text{in}_y}]$ is the input Jones polarization vector, which depends on the azimuth θ and the ellipticity ϵ of the input state of polarization (SOP) [1]. When nonlinear effects are negligible, the output field is given by

$$\boldsymbol{X}_{\text{out}}(\omega) = \boldsymbol{T}(\omega) \cdot X_{\text{in}}(\omega)\boldsymbol{e}_{\text{in}} = X_{\text{out}}(\omega)\boldsymbol{e}_{\text{out}}(\omega) \tag{1}$$

where $X_{\text{in}}(\omega)$ is the Fourier transform of $x_{\text{in}}(t), X_{\text{out}}(\omega) = X_{\text{in}}(\omega)e^{-\alpha z}e^{-j\overline{\beta}(\omega)z}$, and $\boldsymbol{e}_{\text{out}}(\omega) = (\boldsymbol{M}(\omega)\cdot\boldsymbol{e}_{\text{in}})$ is the output Jones polarization vector. All the fiber birefringence effects are included in the frequency dependent vector $\boldsymbol{e}_{\text{out}}(\omega)$, while chromatic dispersion effects are accounted for inside the output scalar field $X_{\text{out}}(\omega)$. The unitary matrix $\boldsymbol{M}(\omega)$ can be written as [1] $\boldsymbol{M}(\omega) = \boldsymbol{R}(\omega)^{-1}\boldsymbol{D}(\omega)\boldsymbol{R}(\omega)$, where

$$\boldsymbol{R}(\omega) = \begin{bmatrix} \cos(k\omega) & \sin(k\omega) \\ -\sin(k\omega) & \cos(k\omega) \end{bmatrix}$$

takes into account the rotation of the PSP's, and the dispersive matrix

$$\boldsymbol{D}(\omega) = \begin{bmatrix} e^{j(\Delta\tau\omega/2)} & 0 \\ 0 & e^{-j(\Delta\tau\omega/2)} \end{bmatrix}$$

takes into account the different propagation speeds on the two PSP's. We have implicitly assumed that the PSP's are aligned with the directions of the $\hat{x}$ and $\hat{y}$ axes and that they rotate in the plane $z = 0$, in the 3-D Poincaré space. This is not restrictive for the following analysis [1]. In the second-order approximation ($\Delta\tau = \Delta\tau_0 + \Delta\tau'\omega$), inverting (1), the time domain output field vector is obtained as

$$\begin{aligned} \boldsymbol{x}_{\text{out}}(t) = {}& \\ \frac{1}{2\sqrt{2}}\Big\{&(a\boldsymbol{u}^* + b\boldsymbol{u})(x^+_{\text{out}}(t+\Delta\tau_0/2) + x^-_{\text{out}}(t-\Delta\tau_0/2)) \\ &+ a\boldsymbol{u}(x^+_{\text{out}}(t-2k+\Delta\tau_0/2) - x^-_{\text{out}}(t-2k-\Delta\tau_0/2)) \\ &+ b\boldsymbol{u}^*(x^+_{\text{out}}(t+2k+\Delta\tau_0/2) - x^-_{\text{out}}(t+2k-\Delta\tau_0/2))\Big\} \end{aligned} \tag{2}$$

where $a = e^{j\theta}\cos(\epsilon + \pi/4)$ and $b = e^{-j\theta}\cos(\epsilon - \pi/4), \boldsymbol{u} = [1, j]$ and $\boldsymbol{u}^* = [1, -j]$, and

$$x_{\text{out}}(t)^{\pm} = \frac{1}{2\pi}\int_{-\infty}^{-\infty}(X_{\text{in}}(\omega)e^{-\alpha z}e^{j\{-\overline{\beta}(\omega)z \pm \Delta\tau'\omega^2/2\}})e^{j\omega t}\,d\omega. \tag{3}$$

Equation (2) states that the effect of a second-order PMD fiber is to generate six filtered replicae of the input pulse for each polarization axis, grouped in three different pairs of replicae, symmetrically centered around the arrival times $t = -2k, 0, 2k$, and weighed by the complex coefficient a or b. If $X_{\text{in}}(\omega) = 1$, (2) represents the exact time impulse response of a second-order PMD fiber. It is possible to extend the method to obtain the exact PMD time impulse response up to any order. Equation (2) gives a clear picture of the different impact of the two second-order PMD coefficients $2k$ and $\Delta\tau'$. In the case $2k = 0$, the output field becomes $\boldsymbol{x}_{\text{out}}(t) = \{x^+_{\text{out}}(t + \Delta\tau_0/2)e_{\text{in}_x} + x^-_{\text{out}}(t - \Delta\tau_0/2)e_{\text{in}_y}\}$, implying that the effect of $\Delta\tau'$ is only to broaden asymmetrically the two replicae centered at $t = -\Delta\tau_0/2$ and $t = \Delta\tau_0/2$ [5]. In the case $\Delta\tau' = 0$, all the six replicae have the same profile, so that (2) can be explicitly calculated with a single Inverse Fourier Transform.

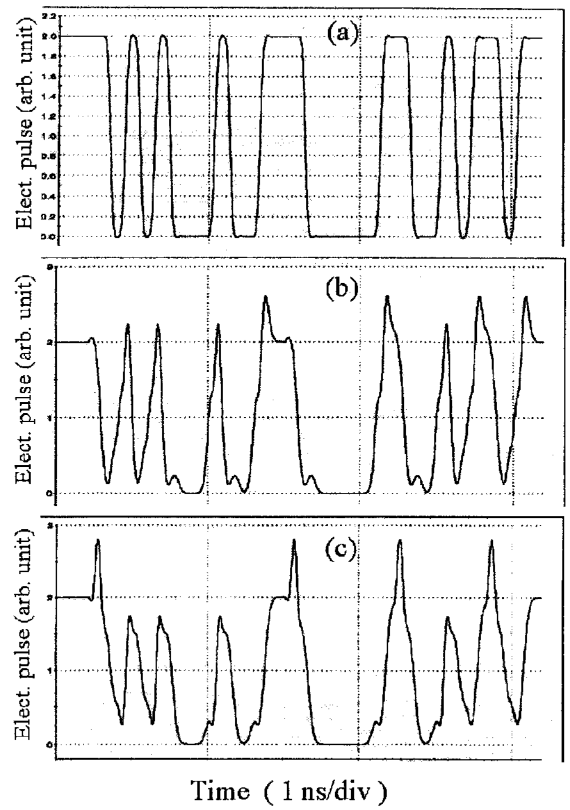

Fig. 1. Numerical results for a PMD fiber with $\Delta\tau_0 = 71$ ps, $2k = 10.8°$/GHz and $\Delta\tau' = 1.3$ ps/GHz. Chromatic dispersion $D_c = 0$. (a) 2^5 bit input sequence of raised cosine pulses. (b) Output sequence when $(\theta, \epsilon) = (0, 0)$. (c) Output sequence when $(\theta, \epsilon) = (\pi/2, 0)$.

Note that we always find energy on both output PSP's, even when only one of the two orthogonal axes is excited at the input. This is actually due to the PSP frequency rotation, which couples the two input orthogonal pulses during the propagation. As an example, suppose that the input SOP is aligned with the $\hat{x}$ axis ($\theta = \epsilon = 0$), that is the fast axis in the above representation, so that $a = b = 1/\sqrt{2}$. The $2k$-induced time delay on the x component of the output field in (2) broadens the pulse centered at $t = -\Delta\tau_0/2$ and avoids a perfect cancellation of the pulse at $t = \Delta\tau_0/2$. Note that, in a first-order approximation, no PSP's coupling takes place, i.e., all the power is predicted to be centered at $t = -\Delta\tau_0/2$. The same imperfect signal cancellation is present on the y component, both at $t = \pm\Delta\tau_0/2$, so that power overshoots are generated on the output sequence, well localized on the edges of the "1" bit sequences. In the next paragraph we will show evidence of this effect both numerically and experimentally, in the case of negligible chromatic dispersion in the fiber. Note that chromatic dispersion, if present, broadens in the same way all six replicae in (2), so that both the pulse broadening and the power overshoots, induced by the $2k$ coefficient, may be enhanced and larger system impairments are expected [1].

IV. Numerical and Experimental Results

Equation (2) is extremely useful, since it gives in closed-form the impact of the second-order PMD on the distortion of any type of input signal. In Fig. 1, we show the calculated

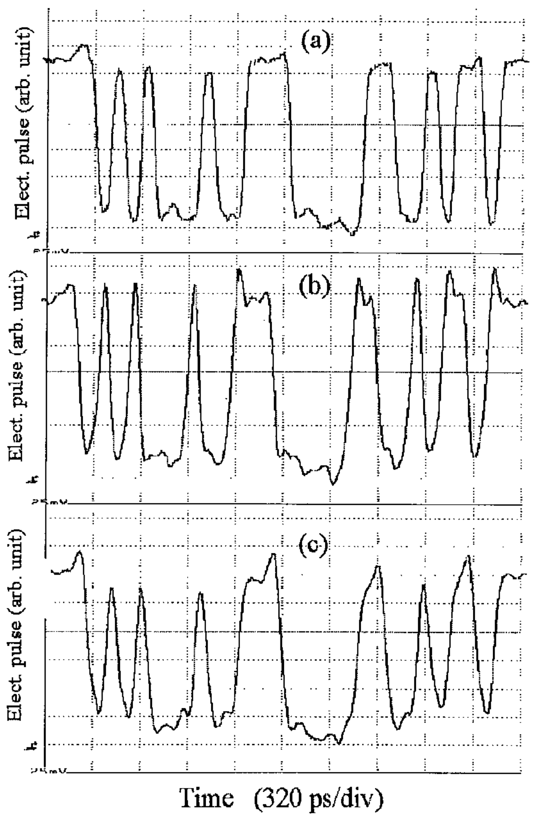

Fig. 2. Same as Fig. 1: Experimental result.

output pulses for a 10 Gb/s transmission in a high PMD fiber with $\Delta\tau_0 = 71$ ps, $2k = 10.8°$/GHz, and $\Delta\tau' = 1.3$ ps/GHz. Chromatic dispersion is set to zero. In Fig. 1(a), we plot the input signal, which is a 2^5 bit sequence of raised cosine pulses. The pulses are chirped by a Mach–Zehnder (MZ) modulator model fitted to the one actually used in the experiment. In Fig. 1(b), we show the output signal when the input SOP is aligned with one of the two PSP's (fast axis). The main effect is due to the $2k$ coefficient, since the value of $\Delta\tau'$ ($\Delta\tau' \simeq 160$ ps/nm, in the same units as chromatic dispersion), is too small to give substantial deformation of the output signal, even if the signal is chirped. The presence of power overshoots on the sequences of "1"s and the presence of energy on the "0"s are due to the imperfect cancellation of the pulses coupled on the orthogonal axis, as explained in the previous paragraph.

In Fig. 1(c), we show the output of the same calculation when the input SOP is aligned with the other axis (slow axis). Pulse distortions are symmetric with respect to the previous case, since now $a = -b$ in (2) and the power coupling between the two components of the output field is reversed. Note that the position of the overshoots on the "1's" reveals which of the two principal axes has been principally excited.

The same type of pulse distortion has been observed in the following experiment. A laser source, externally modulated with the same previous sequence, feeds a high PMD fiber, where the PMD measured coefficients, up to the second order, are the same as the ones used in the numerical calculation. In Fig. 2(a), we plot the modulated transmitter output. The fiber input power of -4 dBm guarantees the operation in the linear regime. In Fig. 2(b) and (c), we plot the two output sequences when the input SOP is aligned along one or the other PSP, respectively. This was obtained by varying the input SOP and searching for the maximum relative delay between the two output sequences, with the minimum distortions on the isolated "1" bits. The power overshoots due to PSP frequency rotation can be clearly observed, both on the "0" and "1" bit sequences, with identical shape and on the same side as those predicted by the theory.

V. Conclusion

In this letter, we give for the first time a closed-form expression for the time impulse response of a linear PMD fiber up to second order. This expression is extremely useful, because it gives a thorough understanding of the different impact of the frequency dependence of the DGD and of the frequency dependence of the PSP's. Besides, it allows a fast estimation of the time pulse deformation on long bit sequences. Experiments have confirmed the results predicted by theory, in particular the second-order induced power overshoots on long sequences of "1's" and "0's."

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